COSMOS IN NUMBERS

The Solar System in 20 Numbers

Twenty checked facts that change how you see the sky

Free guide · NASA-checked

BEFORE YOU START

Real space, real data.

Every number in this guide comes from NASA, JPL or the International Astronomical Union, and each one has its source on the page. Read one a night, then go outside and look up.

The numbers run from the Sun outward, so by the last page you are 25 billion km from home.

NUMBER 01

99.8%

of all the mass in the Solar System is the Sun

Everything else (eight planets, their moons, the asteroids, the comets) shares the remaining 0.2%. Jupiter takes most of that.

Source: NASA, Sun: Facts

NUMBER 02

8 min 20 s

for sunlight to reach Earth

Light covers the 149.6 million km in about 499 seconds. The Sun you see is always the Sun of eight minutes ago.

Source: IAU 2012 Resolution B2 (astronomical unit), speed of light

NUMBER 03

149,597,870,700 m

is one astronomical unit, exactly

The average Earth-Sun distance became a fixed number in 2012. Astronomers measure the Solar System in these units (AU).

Source: IAU 2012 Resolution B2

NUMBER 04

4.6 billion

years: the age of the Solar System

The oldest solid grains found in meteorites formed about 4.57 billion years ago, the closest thing we have to the Solar System's birthday.

Source: NASA, Our Solar System

NUMBER 05

430 °C to -180 °C

the daily swing on Mercury

With almost no atmosphere to hold heat, Mercury bakes in the day and freezes at night: a swing of about 610 degrees.

Source: NASA, Mercury: Facts

NUMBER 06

464 °C

average surface temperature of Venus

Venus is farther from the Sun than Mercury but hotter, because its thick carbon dioxide atmosphere traps the heat. Hot enough to melt lead.

Source: NASA Planetary Fact Sheet

NUMBER 07

243 days

for Venus to spin once

Its year lasts 224.7 Earth days, so one rotation of Venus takes longer than one trip around the Sun. And it spins backwards.

Source: NASA Planetary Fact Sheet

NUMBER 08

107,000 km/h

Earth's speed around the Sun

That is 29.8 km every second. In the time you read this sentence, you have travelled about 100 km through space.

Source: NASA Planetary Fact Sheet (29.78 km/s)

NUMBER 09

384,400 km

average distance to the Moon

When the Moon is near its farthest point (about 405,000 km), all seven other planets would fit side by side in the gap between Earth and the Moon.

Source: NASA Moon Fact Sheet

NUMBER 10

3.8 cm

a year: how fast the Moon drifts away

Laser beams fired at reflectors left by Apollo astronauts measure it. About the speed your fingernails grow.

Source: NASA, Lunar Laser Ranging

NUMBER 11

~22 km

the height of Olympus Mons on Mars

The tallest volcano known in the Solar System rises about 2.5 times higher than Everest stands above sea level (8.85 km).

Source: NASA, Mars: Facts; USGS MOLA data

NUMBER 12

1.8 m

per century: Phobos falling toward Mars

Mars' larger moon spirals inward. In about 50 million years it will crash into Mars or break up into a ring.

Source: NASA, Phobos

NUMBER 13

1,321

Earths would fit inside Jupiter

By volume. Jupiter is more than twice as massive as all the other planets combined.

Source: NASA Planetary Fact Sheet (volume ratio)

NUMBER 14

1.3 Earths

the width of Jupiter's Great Red Spot

In 2017 the storm measured about 16,350 km across. It has been shrinking for over a century, and nobody knows if it will last.

Source: NASA, Jupiter's Great Red Spot

NUMBER 15

0.687 g/cm³

Saturn's average density

Less than water (1 g/cm³). Saturn is the only planet that is, on average, lighter than the same volume of water.

Source: NASA Planetary Fact Sheet

NUMBER 16

10 m

typical thickness of Saturn's main rings

The rings reach up to 282,000 km from the planet, yet the main rings are usually only about 10 m thick. Thinner, for their size, than a sheet of paper.

Source: NASA, Saturn: Facts

NUMBER 17

98°

the tilt of Uranus

Uranus rolls around the Sun on its side. Each pole gets about 42 years of sunlight, then 42 years of darkness.

Source: NASA Planetary Fact Sheet (obliquity 97.8°)

NUMBER 18

2,000+ km/h

winds on Neptune

The fastest winds measured on any planet blow on the farthest one, where sunlight is about 900 times weaker than on Earth.

Source: NASA, Neptune: Facts

NUMBER 19

248 years

for Pluto to go once around the Sun

Discovered in 1930, Pluto will complete its first full orbit since its discovery around the year 2178.

Source: NASA Planetary Fact Sheet (Pluto)

NUMBER 20

~1 light-day

Voyager 1's distance from Earth, late 2026

Launched in 1977, it is more than 25 billion km away. Its radio signal, travelling at the speed of light, now takes about a full day to reach us.

Source: NASA JPL, Voyager mission status

WHAT'S NEXT

Now go outside.

Numbers are better under a real sky. LOOK UP is our 173-page beginner's guide: find the planets and the main constellations with your eyes, another galaxy with binoculars, and every big sky event of 2027, including the total solar eclipse of 2 August 2027. No telescope needed.

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